```
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```

Project : **DATA ANALYSIS ON ZOMATO RESTAURANT AND CUSTOMER BEHAVIOR**

Project Overview

We aim to address the following key problems

1. **Which cuisines are the most popular** → Count how often each cuisine appears and rank them.
2. **Which cities have the highest-rated restaurants** → Average ratings per city to identify top food destinations.
3. **Does online delivery affect ratings** → Compare average ratings for restaurants that do and don't offer online delivery.
4. **What is the distribution of price ranges among restaurants** → Show how many restaurants fall into each price range (1 to 4).
5. **Which restaurants have the most votes** → Identify the top 10 most-reviewed restaurants by vote count.
6. **What is the relationship between average cost and rating** → Use a scatter plot to explore if expensive restaurants get better ratings.
7. **Availability of online delivery and table booking** → Compare average ratings of restaurants with and without table booking.
8. **How many restaurants exist per city in India** → Group by city in India and count the number of listings.

Imported Pandas with allias pd to manage and handle the files with data.

```
import pandas as pd
import numpay as np
```

Mounting Drive to Google Colab with full access to the files.

```
# google colab users

from google.colab import drive
drive.mount('/content/drive')

# /content/drive/MyDrive/Colab Notebooks/zomato.csv
```

```
Mounted at /content/drive
```

Added the dataset

```
import pandas as pd

# Define the file path
file_path = "/content/drive/MyDrive/Dataset/zomato.csv"
```

Zomato csv.....

To show the data from the dataset

```
# Read the CSV file into a DataFrame
zomato_df = pd.read_csv(file_path, encoding='latin1')

# Display the first few records
print("First 5 rows of the dataset:")
print(zomato_df.head())

# Display dataset information
print("\nDataset Information:")
zomato_df.info()
```

```
First 5 rows of the dataset:
   Restaurant ID         Restaurant Name  Country Code              City  \
0        6317637        Le Petit Souffle           162       Makati City
1        6304287        Izakaya Kikufuji           162       Makati City
2        6300002  Heat - Edsa Shangri-La           162  Mandaluyong City
3        6318506                    Ooma           162  Mandaluyong City
4        6314302             Sambo Kojin           162  Mandaluyong City
```

```
                                                Address  \
0  Third Floor, Century City Mall, Kalayaan Avenu...
1  Little Tokyo, 2277 Chino Roces Avenue, Legaspi...
2  Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...
3  Third Floor, Mega Fashion Hall, SM Megamall, O...
4  Third Floor, Mega Atrium, SM Megamall, Ortigas...

                                     Locality  \
0   Century City Mall, Poblacion, Makati City
1  Little Tokyo, Legaspi Village, Makati City
2  Edsa Shangri-La, Ortigas, Mandaluyong City
3      SM Megamall, Ortigas, Mandaluyong City
4      SM Megamall, Ortigas, Mandaluyong City

                                    Locality Verbose   Longitude   Latitude  \
0  Century City Mall, Poblacion, Makati City, Mak...  121.027535  14.565443
1  Little Tokyo, Legaspi Village, Makati City, Ma...  121.014101  14.553708
2  Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...  121.056831  14.581404
3  SM Megamall, Ortigas, Mandaluyong City, Mandal...  121.056475  14.585318
4  SM Megamall, Ortigas, Mandaluyong City, Mandal...  121.057508  14.584450

                           Cuisines  ...          Currency Has Table booking  \
0        French, Japanese, Desserts  ...  Botswana Pula(P)               Yes
1                           Japanese  ...  Botswana Pula(P)               Yes
2  Seafood, Asian, Filipino, Indian  ...  Botswana Pula(P)               Yes
3                   Japanese, Sushi  ...  Botswana Pula(P)                No
4                  Japanese, Korean  ...  Botswana Pula(P)               Yes

  Has Online delivery Is delivering now Switch to order menu Price range  \
0                  No                No                   No           3
1                  No                No                   No           3
2                  No                No                   No           4
3                  No                No                   No           4
4                  No                No                   No           4

   Aggregate rating  Rating color Rating text Votes
0               4.8    Dark Green   Excellent   314
1               4.5    Dark Green   Excellent   591
2               4.4         Green   Very Good   270
3               4.9    Dark Green   Excellent   365
4               4.8    Dark Green   Excellent   229

[5 rows x 21 columns]

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
```

Level 1: Data Understanding & Cleaning

The goal is to understand and clean the dataset before deeper analysis.

We focus on:

Data Overview (Structure, Columns, Data Types)

Checking for Missing Values

Detecting Duplicates

Handling Inconsistencies

To check number of rows and columns

```
zomato_df.shape
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipython-input-1-1441427797.py in <cell line: 0>()
----> 1 zomato_df.shape

NameError: name 'zomato_df' is not defined
```

To check features of the schemas available in the data

```
zomato_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
```

```
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Restaurant ID         9551 non-null   int64
 1   Restaurant Name       9551 non-null   object
 2   Country Code          9551 non-null   int64
 3   City                  9551 non-null   object
 4   Address               9551 non-null   object
 5   Locality              9551 non-null   object
 6   Locality Verbose      9551 non-null   object
 7   Longitude             9551 non-null   float64
 8   Latitude              9551 non-null   float64
 9   Cuisines              9542 non-null   object
 10  Average Cost for two  9551 non-null   int64
 11  Currency              9551 non-null   object
 12  Has Table booking     9551 non-null   object
 13  Has Online delivery   9551 non-null   object
 14  Is delivering now     9551 non-null   object
 15  Switch to order menu  9551 non-null   object
 16  Price range           9551 non-null   int64
 17  Aggregate rating      9551 non-null   float64
 18  Rating color          9551 non-null   object
 19  Rating text           9551 non-null   object
 20  Votes                 9551 non-null   int64
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

To check null values in True and False

```
zomato_df.isnull()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency	Has Table booking	Or deli
0	False	False	False	False	False	False	False	False	False	False	...	False	False	
1	False	False	False	False	False	False	False	False	False	False	...	False	False	
2	False	False	False	False	False	False	False	False	False	False	...	False	False	
3	False	False	False	False	False	False	False	False	False	False	...	False	False	
4	False	False	False	False	False	False	False	False	False	False	...	False	False	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
9546	False	False	False	False	False	False	False	False	False	False	...	False	False	
9547	False	False	False	False	False	False	False	False	False	False	...	False	False	
9548	False	False	False	False	False	False	False	False	False	False	...	False	False	
9549	False	False	False	False	False	False	False	False	False	False	...	False	False	
9550	False	False	False	False	False	False	False	False	False	False	...	False	False	

To check the count of all the null values

```
zomato_df.isnull().sum()
```

	0
Restaurant ID	0
Restaurant Name	0
Country Code	0
City	0
Address	0
Locality	0
Locality Verbose	0
Longitude	0
Latitude	0
Cuisines	9
Average Cost for two	0
Currency	0
Has Table booking	0
Has Online delivery	0
Is delivering now	0
Switch to order menu	0
Price range	0
Aggregate rating	0
Rating color	0
Rating text	0
Votes	0

dtype: int64

To check the column names(default top 5 values)

```
zomato_df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswana Pula(P)
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswana Pula(P)
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswana Pula(P)
					Third Floor, Mega	SM Megamall	SM Megamall,					

To check and drop if any duplicated rows in the dataset

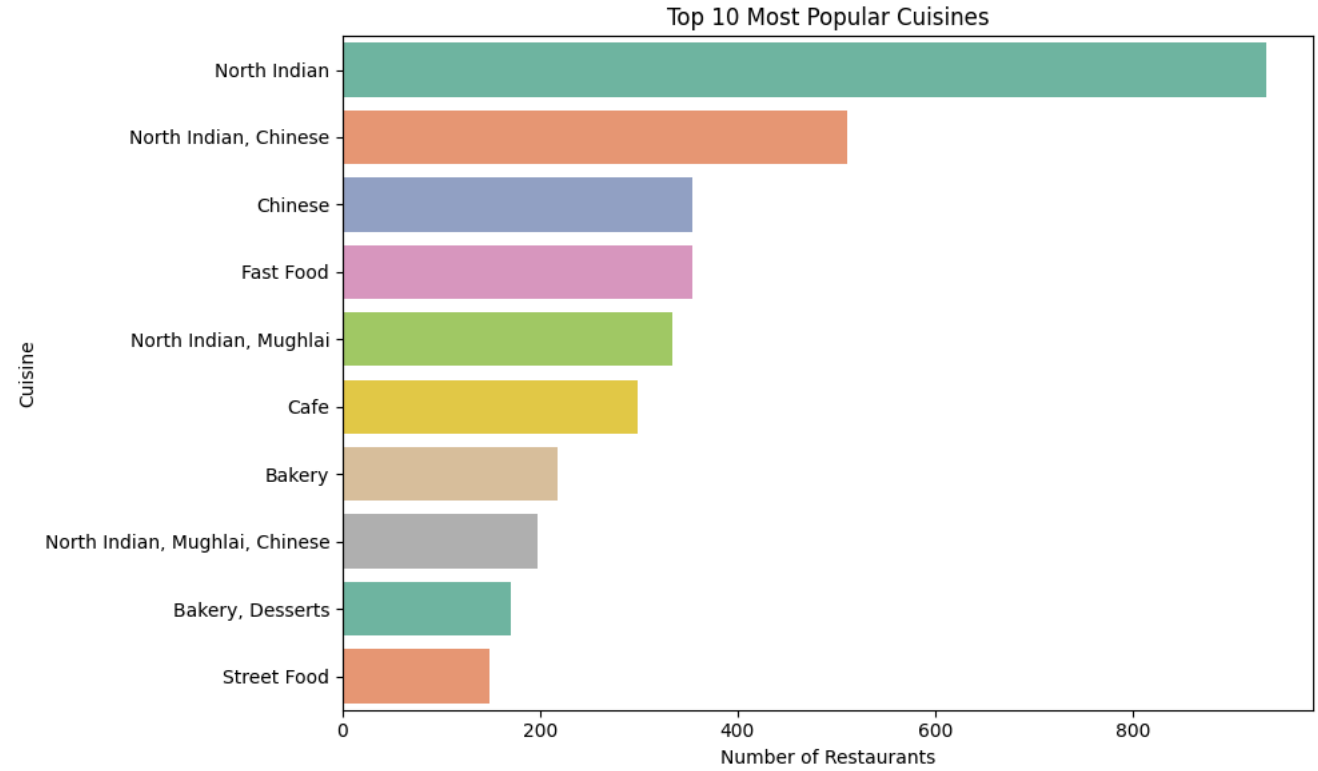
```
duplicated_rows=zomato_df.duplicated().sum()
print(duplicated_rows)
```

```
0
```

1. Most Popular Cuisines

```
# Top 10 cuisines
import matplotlib.pyplot as plt
import seaborn as sns

top_cuisines = zomato_df['Cuisines'].value_counts().head(10)

plt.figure(figsize=(10, 6))
sns.barplot(x=top_cuisines.values, y=top_cuisines.index, palette='Set2')
plt.title("Top 10 Most Popular Cuisines")
plt.xlabel("Number of Restaurants")
plt.ylabel("Cuisine")
plt.tight_layout()
plt.show()
```

```
/tmp/ipython-input-18-3337629528.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `le

  sns.barplot(x=top_cuisines.values, y=top_cuisines.index, palette='Set2')
```

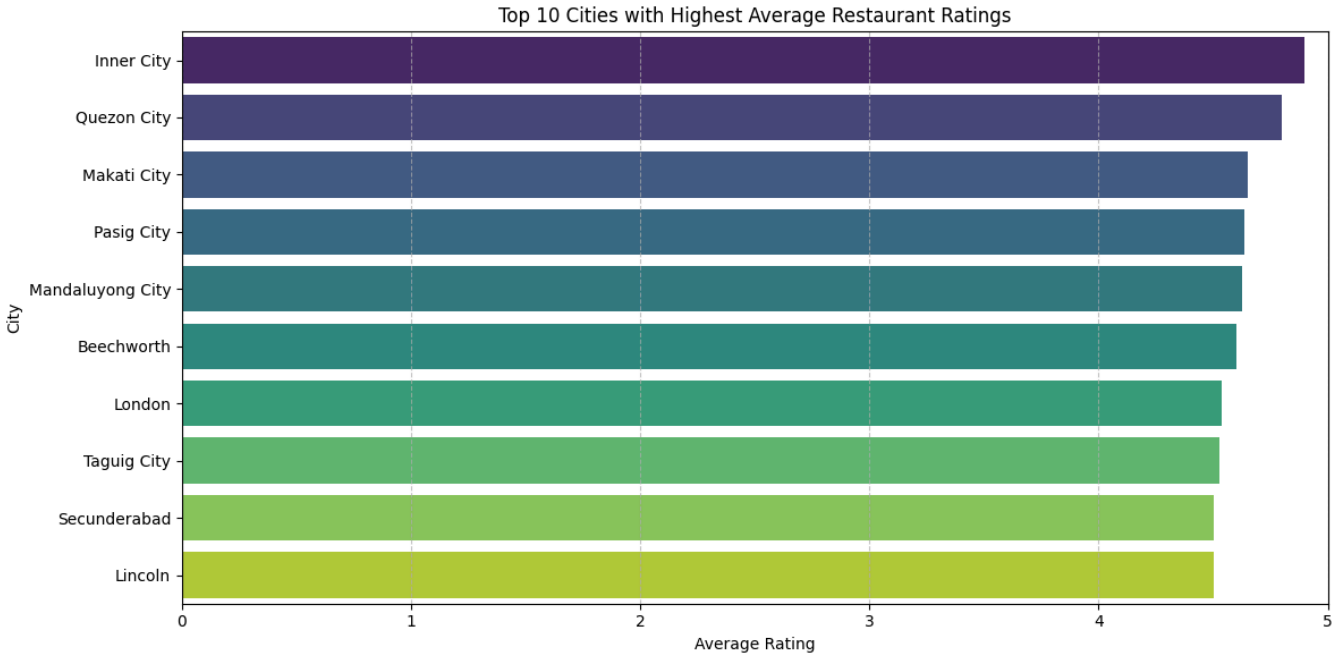


2. Top 10 Highest Rated Restaurants

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv('/content/drive/MyDrive/Colab Notebooks/zomato.csv', encoding='latin1')

# Group by City and calculate the average rating, filtering out 0 ratings (optional)
```

```
city_avg_rating = df[df['Aggregate rating'] > 0].groupby('City')['Aggregate rating'].mean().sort_values(ascending=False).head(10)

# Plotting
plt.figure(figsize=(12, 6))
sns.barplot(x=city_avg_rating.values, y=city_avg_rating.index, palette='viridis')
plt.title("Top 10 Cities with Highest Average Restaurant Ratings")
plt.xlabel("Average Rating")
plt.ylabel("City")
plt.xlim(0, 5)  # Ratings are between 0 and 5
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

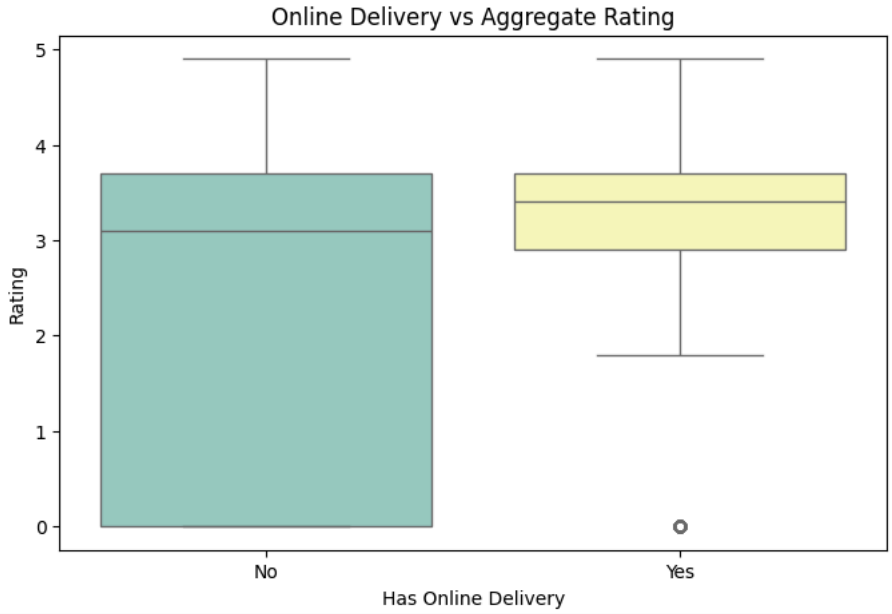
```
/tmp/ipython-input-20-77595702.py:13: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `le

  sns.barplot(x=city_avg_rating.values, y=city_avg_rating.index, palette='viridis')
```



3. Online Delivery vs Rating

```
plt.figure(figsize=(8, 5))
sns.boxplot(x='Has Online delivery', y='Aggregate rating', data=df, palette='Set3')
plt.title("Online Delivery vs Aggregate Rating")
plt.xlabel("Has Online Delivery")
plt.ylabel("Rating")
plt.show()
```

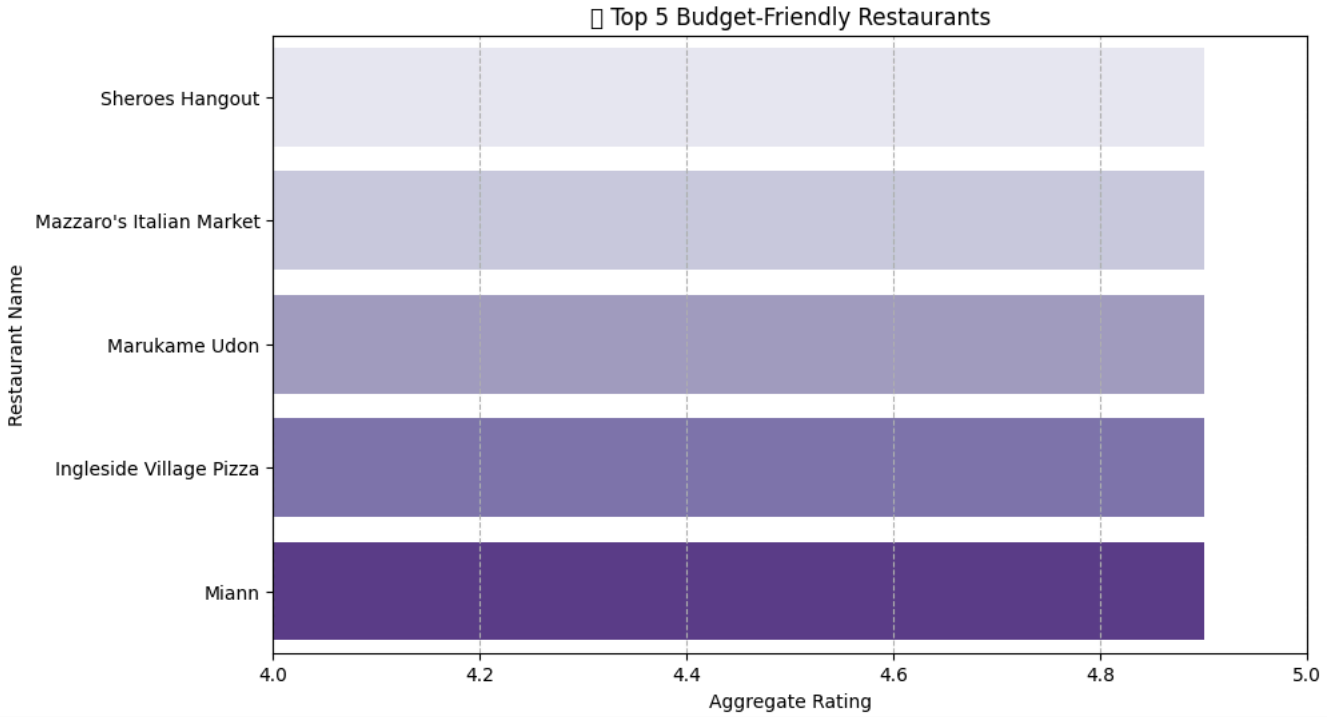
```
/tmp/ipython-input-21-1175005854.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `le

  sns.boxplot(x='Has Online delivery', y='Aggregate rating', data=df, palette='Set3')
```



4. Top 5 Budget-Friendly Restaurants (Price Range = 1 and Rating > 4.0)

```
import matplotlib.pyplot as plt
import seaborn as sns

# Filter: Price range = 1 and rating > 4.0
budget_friendly = df[(df['Price range'] == 1) & (df['Aggregate rating'] > 4.0)]

# Sort by rating and select top 5
top_budget = budget_friendly.sort_values(by='Aggregate rating', ascending=False).head(5)

# Show data
print("💸 Top 5 Budget-Friendly Restaurants:\n")
print(top_budget[['Restaurant Name', 'City', 'Cuisines', 'Aggregate rating', 'Votes']])

# Visualization
plt.figure(figsize=(10, 6))
sns.barplot(x='Aggregate rating', y='Restaurant Name', data=top_budget, palette='Purples')
plt.title("💸 Top 5 Budget-Friendly Restaurants")
plt.xlabel("Aggregate Rating")
plt.ylabel("Restaurant Name")
plt.xlim(4.0, 5.0)
plt.grid(axis='x', linestyle='--')
plt.show()
```

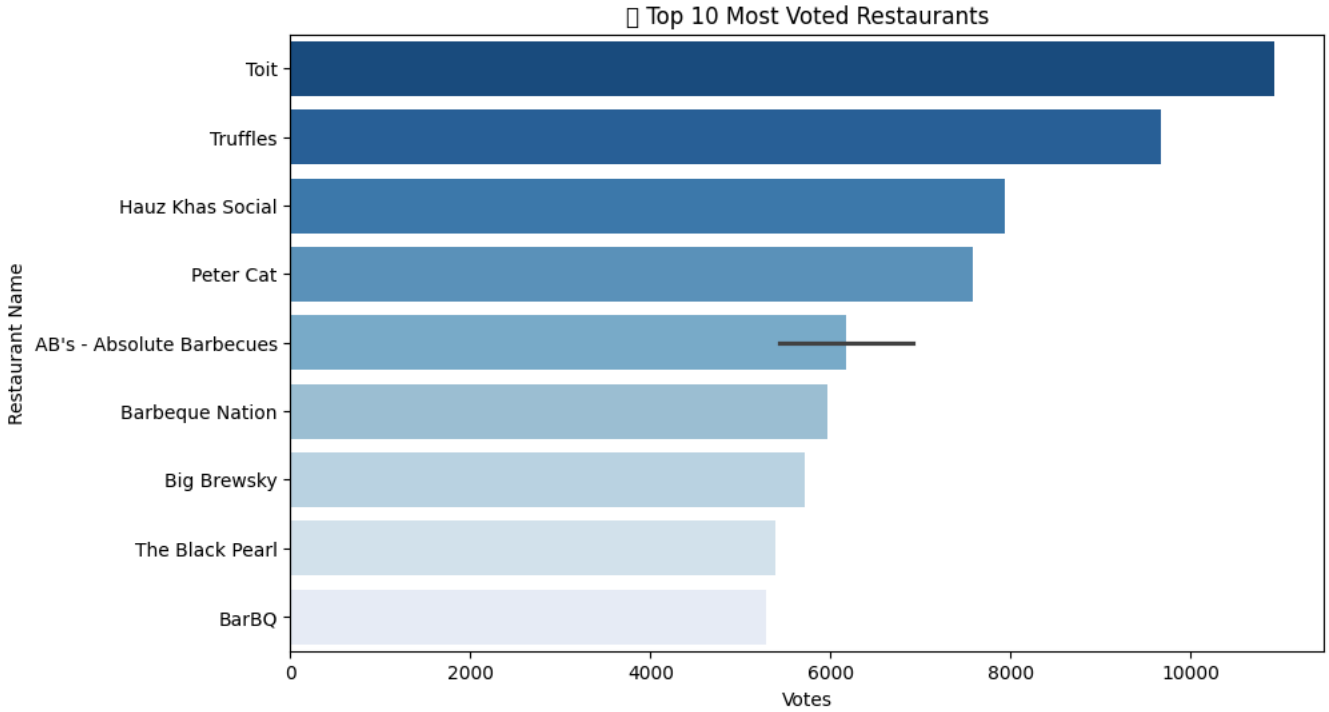
```
💸 Top 5 Budget-Friendly Restaurants:

                Restaurant Name            City                     Cuisines  \
637             Sheroes Hangout            Agra  Cafe, North Indian, Chinese
507   Mazzaro's Italian Market       Tampa Bay                Italian, Deli
429               Marukame Udon  Rest of Hawaii                     Japanese
324    Ingleside Village Pizza           Macon              Pizza, Sandwich
9303                     Miann        Auckland                     Desserts

      Aggregate rating  Votes
637                4.9     77
507                4.9   1424
429                4.9    602
324                4.9    478
9303               4.9    281
/tmp/ipython-input-22-3100609150.py:16: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `le

  sns.barplot(x='Aggregate rating', y='Restaurant Name', data=top_budget, palette='Purples')
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128184 (\N{MONEY WITH WINGS}) missing fro
  fig.canvas.print_figure(bytes_io, **kw)
```



5. Top 10 Most Voted Restaurants:

```
most_voted = df.sort_values(by='Votes', ascending=False).head(10)

plt.figure(figsize=(10, 6))
sns.barplot(x='Votes', y='Restaurant Name', data=most_voted, palette='Blues_r')
plt.title("📌 Top 10 Most Voted Restaurants")
plt.xlabel("Votes")
plt.ylabel("Restaurant Name")
plt.show()
```

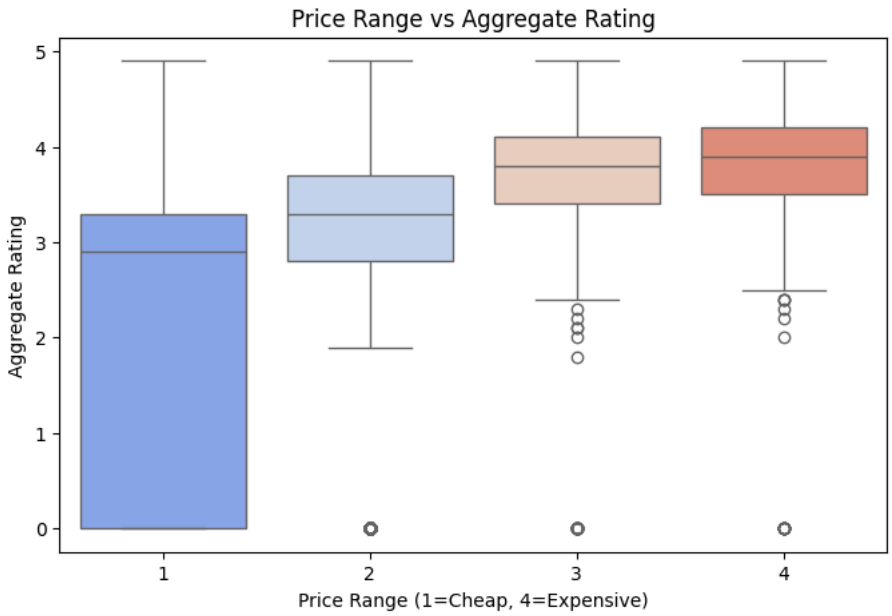
```
/tmp/ipython-input-23-1279578166.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `le

  sns.barplot(x='Votes', y='Restaurant Name', data=most_voted, palette='Blues_r')
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128226 (\N{PUBLIC ADDRESS LOUDSPEAKER}) m
  fig.canvas.print_figure(bytes_io, **kw)
```



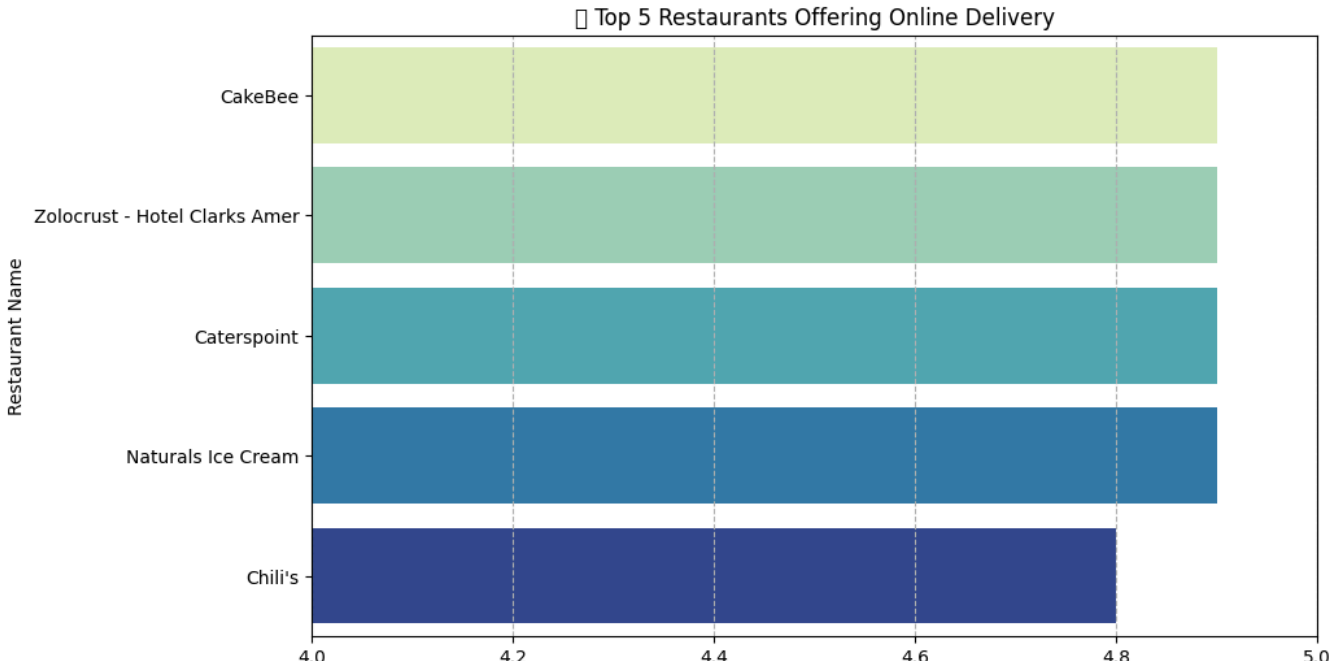
6. Average cost vs aggregate rating

```
plt.figure(figsize=(8, 5))
sns.boxplot(x='Price range', y='Aggregate rating', data=df, palette='coolwarm')
plt.title("Price Range vs Aggregate Rating")
plt.xlabel("Price Range (1=Cheap, 4=Expensive)")
plt.ylabel("Aggregate Rating")
plt.show()
```

```
/tmp/ipython-input-26-3086730220.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `le

  sns.boxplot(x='Price range', y='Aggregate rating', data=df, palette='coolwarm')
```



7. Top 5 Restaurants Offering Online Delivery

```
import matplotlib.pyplot as plt
import seaborn as sns

# Filter: Restaurants that offer online delivery
online_delivery = df[df['Has Online delivery'] == 'Yes']

# Sort by rating and select top 5
top_online_delivery = online_delivery.sort_values(by='Aggregate rating', ascending=False).head(5)

# Print data slice
print("🚚 Top 5 Restaurants Offering Online Delivery:\n")
print(top_online_delivery[['Restaurant Name', 'City', 'Cuisines', 'Aggregate rating', 'Votes']])

# Visualize
plt.figure(figsize=(10, 6))
sns.barplot(x='Aggregate rating', y='Restaurant Name', data=top_online_delivery, palette='YlGnBu')
plt.title("🚚 Top 5 Restaurants Offering Online Delivery")
plt.xlabel("Aggregate Rating")
plt.ylabel("Restaurant Name")
plt.xlim(4.0, 5.0)
plt.grid(axis='x', linestyle='--')
plt.show()
```

```
🚚 Top 5 Restaurants Offering Online Delivery:

                     Restaurant Name        City  \
841                          CakeBee  Coimbatore
2350  Zolocrust - Hotel Clarks Amer      Jaipur
1381                     Caterspoint     Gurgaon
3013             Naturals Ice Cream   New Delhi
809                          Chili's     Chennai

                              Cuisines  Aggregate rating  Votes
841                     Bakery, Desserts               4.9    200
2350        Italian, Bakery, Continental               4.9    322
1381     Mexican, American, Healthy Food               4.9    223
3013                           Ice Cream               4.9   2620
809   Mexican, American, Tex-Mex, Burger               4.8   1262
/tmp/ipython-input-28-78309554.py:16: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `le

  sns.barplot(x='Aggregate rating', y='Restaurant Name', data=top_online_delivery, palette='YlGnBu')
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128666 (\N{DELIVERY TRUCK}) missing from
  fig.canvas.print_figure(bytes_io, **kw)
```